

Cengiz Ozel

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EDUCATION

Georgia Institute of Technology

Master of Science, Computer Science, GPA: 4.0

Atlanta, GA

Aug 2023 – Aug 2025

University of Rochester

Bachelor of Science, Computer Science

Rochester, NY

Aug 2018 – Aug 2022

EXPERIENCE

Ministry of Defense

Software Engineer – Remote

Riyadh, Saudi Arabia

Mar 2025 – Present

- Develop a bilingual Arabic-English conversational AI platform for military healthcare as a Next.js and TypeScript app, powering real-time spoken conversations with humanlike AI avatars.
- Work directly with therapists and speech pathologists at collaborating hospitals, iteratively translating their requirements and feedback into the platform's clinical workflows.
- Design a JSON-driven conversation flow engine that defines each use case (PTSD screening, autism social skills practice, CBT therapist training) as localized states with LLM-classified answer routing.
- Secure the platform with Azure AD login, single sign-on from the national health app, strict validation of all incoming data, session-gated access to every AI and data endpoint, and rate limiting.
- Validated the PTSD screening agent in a 16-participant soldier-clinician IRB study, published at IVA 2026.

Saudi Authority for Data and Artificial Intelligence

AI Engineer – Remote

Riyadh, Saudi Arabia

Nov 2023 – Sep 2024

- Led Flask backend development for an avatar-based AI tutoring platform showcased at the 2024 Global AI Summit.
- Designed the REST API for live tutoring with Azure AI Speech and Unreal Engine avatars over WebRTC, where the avatar speaks each sentence as generated, can be interrupted, and writes code and math on a live whiteboard.
- Built the session analytics pipeline, estimating engagement and heart rate live from webcam frames streamed to the backend and generating post-session quizzes from each conversation.

University of Rochester – Rochester Human-Computer Interaction Lab

Research Assistant

Rochester, NY

Jun 2022 – Nov 2023

- Developed LLM-driven virtual agents with Unreal Engine 5 MetaHuman avatars for communication training, such as end-of-life conversations, iterating on feedback from University of Rochester Medical Center clinicians.
- Developed a Unity-based synthetic data pipeline for hand pose estimation in C# and Python, generating over 500,000 annotated hand images within 24 hours, with OpenCV augmentation and CSV to COCO JSON conversion.
- Trained HRNet and ViTPose models on the synthetic dataset with PyTorch, reaching 96.08% PCK near state-of-the-art performance, with Faster R-CNN and YOLOv7 hand detectors as a preprocessing stage.

PROJECTS

SignalBerry github.com/cengizozel/SignalBerry

Sep 2025 – Jul 2026

- Built a native Signal messenger client for BlackBerry 10 in Java, with a self-hosted Python and Flask bridge that persists every message, reaction, and receipt to SQLite so offline clients catch up on everything they missed.
- Added TLS 1.3 to Android 4.3 via Conscrypt, kept uploads at constant memory on the small-heap device, and shipped the stack with Docker behind a bearer-token nginx proxy, verified by 54 pytest tests.

PUBLICATIONS

- Design and Evaluation of a Culturally Adapted Multimodal Virtual Agent for PTSD Screening. **C. Ozel** et al., **IVA 2026**
- Hi5: Synthetic Data for Inclusive, Robust, Hand Pose Estimation. M. Hasan, **C. Ozel**, et al., **ACII 2025**
- SAPIEN: Affective Virtual Agents Powered by Large Language Models. M. Hasan, **C. Ozel**, et al., **ACIIW 2023**

SKILLS

- **Languages & Databases:** Python, TypeScript, SQL, C++, C#, Java, Bash, MATLAB, PostgreSQL, SQLite
- **Frameworks & Libraries:** Next.js, Flask, React, Node.js, Tailwind CSS, Playwright, Vitest
- **Cloud & DevOps:** Alibaba Cloud (ApsaraDB), GCP (Vertex AI), Azure (AD), AWS, Docker, GitHub Actions